

Q1) Depending on the following snapshot of memory, answer the following questions (6pts)

emulator: noname.com			
file math debug view external virtual devices virtua			
Load reload step back ID single step			
registers			
	H	L	
AX	00	47	07110: 2A 042
BX	00	00	07111: 23 035
CX	00	00	07112: 40 064
DX	00	00	07113: 25 037
CS	0700		07114: 4D 072
IP	010F		07115: 4B 075
SS	0700		07116: 90 144
SP	FFFE		07117: 90 144
BP	0000		07118: 90 144
SI	0118		07119: 90 144
			0711A: 90 144
			0711B: 90 144
			0711C: 90 144
			0711D: 90 144
			0711E: 90 144

a) What is the physical address for the memory location pointed by SI?

07118H

b) What is the ASCII code for symbol '@' as it shown in figure?

40

c) What is the DS value?

0700H

d) Write the suitable instructions to swap between memory locations 07110H and 07111H?

~~mov ax, 07110H~~
~~mov si, 07110H~~

mov ax, 700h
mov ds, ax
mov si, 110h
mov cx, esi
inc si
mov bx, esi
exchg cx, bx

Q2) Trace the following (10 pts)

a- org 100h
mov di, 00
mov ax, 54D
mov bl, 2
L3: cmp al, 10
jb L
div bl
add ah, 30h
mov x[di], ah
inc di
mov ah, 0
jmp L3
L: add al, 30h
mov x[di], al
hlt

ah = 30h

36h
x[3] = 36

x db 3 dup(?)

al = 36h, ah = 09h
x = 30h, 31h, 31h, 36h

mov al, 1
mov ch, 0
mov dx, 20h
mov cl, 2
L: add dx, cl
mul dl
loop L

Dx = h
Cl = 0

Ax =

MOV DL, 0
MOV CX, 3
MOV AL, 5
L3: SHR CL, 1
JNC L1
ADD DL, AL
L1: SHL AL, 1
CMP CX, 0
JE L2
JMP L3
L2:

AL = 14h
20D

d-

MOV AX, 8000H
MOV BX, 0001H
SUB AX, BX

Ax = K
Cf = 0
Sf = 0
Pf = 1

DL = 0FH

Student Name:

Q3) Write a program to find how many numbers exist in the following array 'A1' that are less than or equal 6 then store the result in Y (8 pts)

A1 DB 3, 7, 8, 5, 21, 6

```
org 100h
mov cx, 6
mov si, offset A1
mov bl, 0
L2: mov ah, [si]
    cmp ah, 6
    jbe L1
    inc si
    jmp L2
L1: inc bl
    mov y, bl
    hlt
A1 DB 3, 7, 8, 5, 21, 6
Y DB ?
```

Q4) Write an equivalent Assembly code for the following equation:

$res = ((X+10) * Y)/Z$ (6 pts.)

Note: X, Y and Z are byte sizes (USING INDIRECT ADDRESSING MODE TO ACCESS THE DATA)

```
org 100h
mov si, 0
mov al, X[si]
mov bl, X[si]
inc si
mov cl, X[si]
ADD al, 10
mul bl
Div AX, cl
mov res, AX
```

Faculty of engineering technology
Computer Engineering department



final assembly lab Exam(A)

2nd Semester 2017/2018

Student No:
Student Name:

INSTRUCTOR NAME: Mervat el rawashdeh
Muhammad el waher

Exam time: 50 min

➤ Answer the following Question :

Question # 1: Write an assembly code to input number (from 0 to 9) from keyboard then print "even" if number is even ,other wise print "odd" Write complete program? (10 pts)

Question #2: Trace the following (10 pts):

#1:

Org 100h

Mov ah,9

Lea dx,str

Int 21 h

Hlt

Str db " assembly",09,41h,42h,24h

#2:

org 100

Mov ah,9

Lea dx,x1

int 21H

hlt

X1 db "ab"

X2 db "cd5"

X3 db "ef5"

81.

MOV AX, 3232h

MOV BX, 2734h

PUSH AX

PUSH BX

POP CX

POP BX

KCHG

AX=

CX=

BX=

Question 82

Choose the correct answer for following points: (14pts)

1. Assuming AL = 4 and BL = 5, the code:
MOV AL, BL
MOV BL, AL
is equivalent to:

- a) MOV BL, AL
MOV AL, BL
- c) MOV BL, 4
- d) MOV BL, 5
- e) KCHG AL, BL

2. Assuming AL = 1 and BL = -1, the instruction: MUL BL will produce:

- a) AX = 1
- b) AX = -1
- c) AX = 00FFh
- d) AX = FF00h
- e) AX = 0100h

3. Assuming AL = -1 and BL = -1, the instruction: DIV BL will produce:

- a) AX = 1
- b) AX = -1
- c) AX = 00FFh
- d) AX = FF00h
- e) AX = 0100h

Faculty of engineering technology
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final assembly lab Exam(A)

Student No:

Student Name:



2nd Semester 2017/2018

Instruction name: Mounir el ramessy el
Mohammed el sarak

Exam time: 50 min

4. neg is equivalent to :

- a) multi -1 b) Subtraction c) div d) add e) a+b

5. LEA is means

- a) Load effective address b) Load address. c) Load physical address. d) None of them .

6. Push is means:

- a) Operand onto the top of the hardware supported register
b) places its operand onto the top of the hardware supported stack in memory
c) Operand onto the top of the hardware supported stack in memory
d) None above

7. test instruction is equivalent to:

- a) cmp b) jnz c) shl d) shr e) all of them

Question #4:

Write an assembly code to exchange the values in the two arrays using stack (push and pop instructions)? (10)

Question #5:

Suppose that DS=0200H, BX=0300H, and DI=400H. Determine the memory address accessed by each of the following instructions, assuming real mode operation(6pts)

(a) MOV AL, [1234H]

(b) MOV AX, [BX]

(c) MOV [DI], AL

The End

Computer Engineering Department
Assembly Lab Midterm Exam

Date:

Student Name:

Question: Write a program that finds the count of negative even numbers in an array of 10 elements, and stores it in memory location **count**.

Note: an example is:

$ARR[] = \{2, 9, -1, 0, -5, 3, 8, -4, -2, 7\}$

$count = 4$

Tuesday 11:00 - 14:00

صباح جميل نأهني اساميل

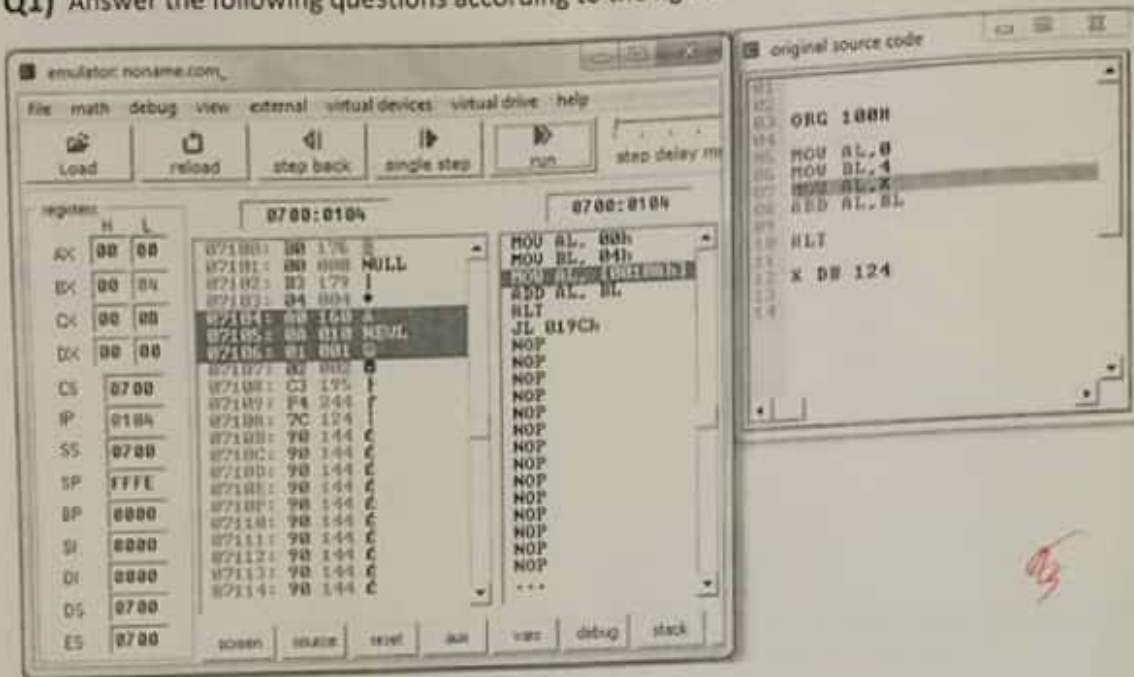
Al-Balqa Applied University
Computer Engineering Department
Student Name:

Microprocessor Systems Lab
Mid Exam (30%)

Faculty of Engineering Technology

Instructor: Eng. Nawal Alzabin

Q1) Answer the following questions according to the figure below



28
30

A) What is the physical address of the variable X? (1 Marks)

07104H

B) What is the value of IP register after executing the highlighted instruction? (1 Marks)

107H

C) If the machine code of the HLT instruction is F4H, what is the logical address of the HLT instruction? (1 Marks)

109H

D) Fill the value of AL register and Flags after executing the instruction "ADD AL, BL". (2 Marks)

AL = 80H

CF = 0, PF = 0, AF = 1, ZF = 0, SF = 1, OF = 1

Q2) (5 Marks) Read the word at address DS:600H, shift its value right and store it back to memory.

~~Org 100H
Mov SI, DS:600H
Mov AX, [ESI]
SHR AX, 1
Mov [ESI], AX
Hlt~~

Org 100H
LEA SI, DS:600H
Mov AX, [ESI]
SHR AX, 1
Mov [ESI], AX
Hlt

6

Q4) Write an assembly

1- If the character is 'a' then print your name on screen as follow:

9

2- If the character is 'b' store it in variable W.

```

Mov DX, 900 H
Mov DS, DX

```

MOV AL, [SI]
L1: MUL BL → L1: Div BL ✓
CMP AL, 0 → Cmp AH, 0 ✓
JNE L2 → JE L2
MOV [SI], 0
L2: → Inc SI ✓
LOOP L1

Q.3) Org 100H
LEA SI, X
LEA DI, X+1
Mov CX, 4
L1:

Mov AL, [SI]
Mov BL, [DI]
Mov [DI], AL
Mov [SI], BL
Add SI, 2
Add DI, 2
Loop L1
H/E

X DB 1, 2, 3, 4, 5, 6, 7, 8

Q.4) Org 100H

Mov CX, 5
Mov AH, 1
Int 21H
Cmp AL, 61H
JE L1
Cmp AL, 62H
JNE L2
Mov W, AL
Jmp L2
L1:
Mov AH, 2
Mov DX, 08H
Int 21H
Mov AH, 9
Mov DX, offset X
Int 21H
Mov AH, 2
Mov DL, 09H
Int 21H
Mov AH, 9
Mov DX, offset X
Int 21H

Mov AH, 2
Mov DL, 0AH
~~Int 21H~~
Int 21H
Mov DL, 0DH
Int 21H
Loop L1
L2:
HLT
W DB ?
X DB 'ahmad f'

Student Name: _____

Q3) Trace the following (10 pts)

```
x dw 4020h, 22d4h, 0aaffh
y db 7, 9, 5, 1
mov bx, offset x
push [bx+1]
mov ax, [bx]
add bx, 2
mov cx, [bx]
mov si, offset y
sub si, 1
push [si]
pop bx
pop dx
```

```
mov di, 0
mov cx, 3
mov al, 5
l3: shr cl, 1
jnc l1
add di, al
l1: shl al, 1
jcxz l2
jmp l3
l2:
```

al = 5 cl = 4

```
array dw 2, 4, 6, 8
count dw 5
mov ax, 0
stc ; set cf=1
mov cx, count
mov si, offset array
label1: adc ax, [si]
add si, 2
loop label1
label2:
```

ax = 1A cx = 0

ax = 4020h
cx = 22d4h
dx = d44h

(5)

Q4) Write an assembly program that counts number of (m) letter in the following string (8 pts)
str DB 'assembly programming lang.'

```
org 100h
LEA SI, str
mov cx, 26
mov AL, 0
L1: CMP [SI], 'm'
Jnz L2
Inc AL
L2: LOOP L1
HLT
```

(8) 7

~~str~~ str DB 'assembly programming lang.'

Q1:- X dB 2, y dB -8, z dB 9 write a program that finds the maximum number and stores it in memory location "maximum"

→ Solution:-

X dB 2

y dB -8

z dB 9

maximum dB ?

mov maximum, x

cmp maximum, y

JBE L1

cmp maximum, z

JBE L2

JMP L3

L1: mov maximum, y

cmp maximum, z

JBE L4

JMP L3

L4:

L2: mov maximum, z

L3: HLT

Q2:- Array dB 1, 2, -4, 9, -8 write a program that converts negative numbers to positive numbers in array

→ Solution:-

mov si, 0

mov cx, 5

L3: test array[si], 80H

JNZ L1

Inc si

JMP L2

L1: Neg array[si]

Inc si

L2:

loop L3

HLT

array dB 1, 2, -4, 9, -8

Q3:- Using logical gates write a program to find $2*N$ where N dB 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

→ Solution:-

mov si, 0

L1: SHL N[si]

Inc si

JMP L1

HLT

N dB 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Q4:- write an equivalent program to do the following:-

a) mov cx, bx

mov bx, ax

mov ax, cx

→ solution:-

XChg ax, bx

b) Loopne

→ solution:-



mov cx, 10.

L2:

Dec cx

cmp cx, 0

JNZ L1

JMP L3

L1: JNC L2

L3: HLT

c) neg ah

→ solution

Not ah

Inc ah.

Q5:- write a program that converts capital letters to small letters and vice versa in:- array dB 'A', 'b', 'C', 'D', 'e', 'F'

→ solution:-

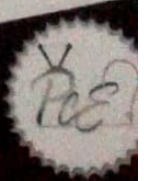
mov si, 0

L3: test array[si], 20H

JNZ L1

OR Array[si], 20H

Don't miss the change..
Be part of it !!



JMP L2

L1:- And array [si], 0DF

L2: Inc si

Loop L3

HLT

Array DB 'A', 'b', 'c', 'D', 'e', 'f'

Q6:- $X=25$, $y=3$ $al = x/y$ $ah = X \cdot y$ write this program
without using DIV or IDIV

→ Solution:-

mov cl, 1

L3: mov AL, X

mul cl

cmp AL, y

JBE L1

Dec cl

mov AL, X

mul cl

JMP L2

L1: Inc cl



JMP L3

L2: mov AL, CL

sub y, AL

mov AH, y

HLT

X dB 25

y dB 3

Q7: convert the following C code to assembly code, where X and y are bytes (X=4, y=6). IF (X >= 5) X = Y/3
else X = 2 - Y

→ Solution

